Sean W. Evans

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Summary

Systems engineer with production experience in machine learning, high-throughput data processing, and software infrastructure, now focused on research engineering in high-performance mathematical computing, compilers, GPU/FPGA acceleration, and database internals. Builds experimental systems from mathematical formulation through low-level implementation, validation, and performance measurement using C++, C, CUDA, Rust, Python, SQL, LLVM, and RTL.

Experience

Data Conversion Laboratory **Remote**
Machine Learning Engineer *May 2021 – Feb 2025*

- Designed a Detectron2-based document segmentation pipeline that reduced manual classification by 50%.
- Built a CNN–RNN LaTeX OCR system for high-accuracy equation and document recognition.
- Automated document styling with NLP and the Microsoft Office SDK, reducing a 40-hour workflow to 2 hours.
- Implemented an OpenCV-based checkbox detector achieving greater than 96% accuracy across thousands of formats.
- Developed high-throughput processing pipelines capable of handling millions of pages per week.

Data Conversion Laboratory **Remote**
Software Engineer *Sep 2020 – May 2021*

- Architected a PDF cleaning and OCR preprocessing system scaling to more than 100,000 pages per week per server.
- Automated LaTeX and JATS XML correction workflows, replacing a full-time manual process.
- Developed DOCX-to-XML conversion tooling using Microsoft Office Interop and C#.

Data Conversion Laboratory **Queens, NY**
Lead Technology Analyst *Sep 2019 – Sep 2020*

- Led workflow optimization efforts, mentored junior engineers, and coordinated technical requirements with stakeholders.

Data Conversion Laboratory **Queens, NY**
Technology Analyst *May 2018 – Sep 2019*

- Developed custom data-conversion tools for production support and maintained high-volume enterprise pipelines.

Selected Research & Engineering Projects

- **pynq_butterfly** — Exact FPGA accelerator for OpenFHE BGVRNS ciphertext multiplication and BV relinearization on a PYNQ-Z2. Achieved 245.61 exact relinearized ciphertexts/sec at 98.84% of the calculated transport ceiling; validated 6,291,456 residue outputs against OpenFHE with zero mismatches.
- **Lockstep** — Data-oriented systems programming language and compiler for deterministic, high-throughput compute pipelines. Implements straight-line SIMD execution, static memory topology, semantic type checking, LLVM IR generation, manual vector lowering, benchmarking, and LSP support.
- **fluid-sims** — CUDA-accelerated numerical simulation laboratory spanning SPH fluids, hypersonic flow, reaction-diffusion, Burgers flow, shallow-water systems, and 3-D fluid dynamics.
- **pg_gpt2** — Complete GPT-2 implementation inside PostgreSQL, including a native C tensor engine, relational reverse-mode autodiff, AdamW optimization, checkpointing, tokenization, training, and inference through SQL.
- **WarpDB** — GPU analytical query engine that parses SQL-like expressions into ASTs and JIT-compiles them into CUDA kernels using NVRTC, with columnar-data and multi-GPU execution support.

Technical Skills

Languages	Rust, C++, C, NASM, CUDA, Python, SQL, TypeScript
Systems	Linux, GPU kernels, FPGA/RTL, LLVM, SIMD, multiprocessing, distributed systems, consensus algorithms, compiler tooling, AST analysis
Databases	PostgreSQL internals, extension development, query execution, vector search, ETL pipelines, XML processing
ML / CV	PyTorch, Detectron2, OpenCV, OCR pipelines, neural-network training and inference

Education

Long Island University **January 2026**
B.S. in Mathematics